Knowledge Infrastructures and the Ontology Pipeline for AI Systems with Jessica Talisman

So I'm here to present about the ontology pipeline. Some of you may have heard of the ontology pipeline. It is a framework that I created essentially to lend context and semantics to our data ecosystems.

So Matt was right. I've been evangelizing and talking about ontology pipeline for quite some time conceptually. Ontology pipeline, if any of you have heard of the semantic web is a riff on the semantic web's concept of semantic expressivity and the semantic spectrum.

So the idea is iterative stages to build semantic expressiveness starting from a pretty primitive representation all the way to a fully expressive context rich knowledge ecosystem. So this ontology pipeline is essentially a framework for context engineering. And we've all heard context engineering.

In fact, Lack and Lisa both alluded to that, the idea of context windows and RAG and different opportunities for presenting context. But the idea is that we work through these stages of building semantic structures. One is used and leveraged to build the next stage.

So control vocabulary, metadata standards, taxonomy, the source, ontology. And with the combination of all of those elements, we end up with an RDF knowledge graph, different from a labeled property graph. The two are siblings within the graph space.

And the idea is introducing RDF or resource descriptive graphs that are rich with ontologies that give our knowledge graph essentially a semantic backbone and meaning and context through relationships and the act of describing things. So these structures from controlled vocabulary all the way through knowledge graph are encoded within systems and are meant to be machine readable and interoperable at scale. The ontology pipeline is not restricted or constrained by tables, by primary keys and foreign keys.

We break out of the table constraints in order to have the level of descriptiveness and expressiveness that we need to really describe and lend context to machines. So as we know, building semantic knowledge infrastructures can be messy. And this has already been touched upon by our speakers.

So part of this is that the most recent AI wave, which began about four years ago, came in an opportune time because organizations were drowning in lots of data. The name of the game since the 90s really has been big data. And we've all been involved in it in some form or context.

And so part of the big data movement in the 90s was how do we maximize, how do we optimize storage? The focus was on storage first and query layers second. And so in managing big data, that has been one of the AI waves. And what's interesting is that that exact same moment in the 1990s as we were dealing with big data and sort of the insurgence of that, the library and information science space was working on creating rich metadata ecosystems that were threaded through the machine readable code for governance, reliability, consistency, basically the FAIR principles.

If any of you heard of FAIR, it's findable, accessible, interoperable, retrievable. That was the primary focus in library and information science spaces because library and information science spaces were one of the first sort of document and resource-based ecosystems to come online starting in the 1960s, well before the common internet or what we know as the web came about in the 90s. So libraries were used to having to create these large worldwide interoperable catalogs that were searchable and where records can be reconciled and retrievable.

And a lot of those same systems have been maintained and still exist today connecting worldwide libraries for lending and also for access to information and knowledge. And as we know also, this is complicated. It's not easy.

I think Lisa mentioned about retrofitting that we're kind of in this stage where there's a lot of retrofitting going on. And so it's complicated because we're all trying to solve the problem of how do we introduce context? How do we work well with AI? It's really about human machine collaboration. And so in working through these problems, it's also making us examine our skill sets and look at what we have to add to our toolbox in order to optimize and be most efficient while still adding context and focusing on data quality.

Data quality being the main component. So I don't know if any of you have heard of the Ikea effect. It's not necessarily a good thing in most contexts.

So the Ikea effect is that idea. It's actually a cognitive bias where we tend to love things more if we've made it. And building data and information ecosystems and infrastructure is intense.

But it's in the act of owning these infrastructures and being intimately part of the processes that we tend to invest ourselves more in the continued iterations and development of these ecosystems. And the same thing applies within semantic ecosystems. So when I say semantic, I don't want you to get confused with the semantic layer, although that is a component.

We are talking about semantic relevant to the semantic web where we have these rich knowledge graphs, ontologies, taxonomies, controlled vocabularies, these expressive semantics where we are able to translate and communicate the meaning of things within our ecosystems. So by building skills and learning a little bit outside of the norm or of the normal tool sets and toolboxes, we start to learn how to do these things ourselves, which is critical to understanding context for machines. Essentially, what we are doing with the ontology pipeline is introducing symbolic AI.

The current form of AI that we are working with is generative AI, which is different than symbolic

AI. Generative AI is mainly statistical. The offset or the balance, the counterbalance to our generative systems is introducing symbolic AI, which gives the rich context, which is the context engineering so desperately needed for AI implementations to drive accuracy and reliability.

So the idea is to sort of capitalize on the IKEA effect, give people, empower people to learn how to build these ecosystems and infrastructures for knowledge and context so that we have more at our disposal in terms of being able to leverage different tool sets for different jobs to lend context and to build even, for example, RAG systems for our AI implementations. So the first step in ontology pipeline is a controlled vocabulary. And in case, you know, at the end I'll have a QR code to my sub stack, I write about this pretty extensively, including how to build these systems.

But the first step is controlled vocabulary. And broadly, controlled vocabulary means that we have a determined set of terms that become concepts and form into concepts using a light, light, light touch ontology and domain model or data model called simple knowledge organization system, SCOS. So you'll hear SCOS dropped here and there.

But it basically forms concepts and also allows us with the concepts to introduce alternative labels, which are also commonly referred to as aliases. So a concept basically wears a backpack, and I know this is a cheesy way to describe it, but in the backpack, it's kind of like you have a concept will have an alternative label or multiple alternative labels that will have hidden labels, the definition notations. You imbue a concept with all of the rich context.

So it's not simply just lending context. The concept itself has context and meaning and definition and disambiguation. So the control vocabulary layer is the first step because how do we know what to structure and model if we don't have definitions? We need to deal with those fundamental aspects of our data so that, so that we can drive reliability and accuracy.

So with this, there's also another component of building these controlled vocabularies, and I encourage you to look at ONSI Z39.19. It's a guideline for construction and format and management of monolingual controlled vocabularies. There is a rule base that goes to building concept models. So this is really about conceptual modeling.

The reason we do this is our goals are to eliminate ambiguity, synonym control, testing and validation, and that we can start to establish relationships between terms. This is how we begin to model context. So these things are standardized and there are rule bases.

SCOSS is what helps to give it structure so that it is machine readable and understandable. That is absolutely essential to understand about this process. The guideline for the construction of these vocabularies is more or less pretty easy to understand, but I definitely recommend that you use this in partnership alongside your SCOSS W3C standard documentation.

So the idea, and this is actually, I pulled this from NASA's Thesaurus, and the Thesaurus is a form of a controlled vocabulary. But in this sense, this is a controlled vocabulary where we can see that there are indicators, and these are really important. So there's some flexibility in how we construct controlled vocabularies, but specific to NASA, they have decided on a general term that reaches across multiple catalogs, and that is its indicator to tell us that this is a general term used across the entire vocabulary space, and then you have related terms.

So although it's a flat list, I want you to think of this and transpose this to what an index in a recipe book looks like. So this is a way that we can really, really help a machine to understand context because we encode this very simply in this sort of Thesaurus-like structure, but essentially this is our controlled vocabulary. The idea is that we're representing words that then we transform into official concepts, and so these indicators become crucial to starting to formulate what our data means, what our concepts mean, applying definitions and consistency across the entire vocabulary.

Our next step is metadata schemas, and I'm so glad that Lisa went right before me because I agree with her very, very, very passionately about metadata schemas. Metadata schemas are a great source for discovering vocabulary terms to consider and to work into your framework for deciding on what a primary label might be, but they also highlight how our data infrastructure are structured, organized, and shared. So this also becomes the critical interoperability layer as we know and many of us have dealt with is that interoperability is key to successful AI implementations, and so the metadata schema is critical to understand early so that we keep that in mind as we design the different phases of our semantic ontology pipeline.

So one thing I also want to point out is that with the controlled vocabulary stage, once the controlled vocabulary stage is ready and you've completed it, you could actually test and use the controlled vocabulary as is, and so each of these phases in the pipeline are usable. It's not that you have to wait till the very end, and for many organizations, you may have a controlled vocabulary, arrive at metadata schemas, and that's all you're going to work with for right now. It's just enough context for many as we start to build the levels of semantic maturity and complexity.

And this is why I really like the idea of us really collaborating and understanding that metadata is the love language of the enterprise. If there's anything across different and varying types of infrastructures, machine readable code, different ways of representing data, metadata is where we can have sort of a common understanding where we can apply transformers, where we can have schema reconciliation and validation layers. This is critical to understand because there is a little bit of mental gymnastics in terms of starting to decode and design and architect interoperability layers between our semantics, so semantic expressions and semantic ecosystems, and our relational database structures.

And this goes for ETL and ELT. How are we going to transport, reconcile, and transform? And so metadata becomes the love language of the enterprise. It's a place that we can work and collaborate to, and I think that this is where the new waves and innovation is going to happen in AI is really about integrations and designing sort of the new future of how we're going to transport data and reconcile data in order to deliver really rich context and meaning to AI systems.

The next stage after we, you know, have examined and considered and tested our controlled vocabulary and metadata is that we're looking at taxonomies. Taxonomies, many of you may be familiar with, and I know it's been mentioned at least in Lach's presentation, is that they organize vocabularies and metadata as hierarchies. These are parent-child relations.

I would warn you that, you know, there's the concept of faceted taxonomies, which is just a basic two-level structure, and when I talk about taxonomy, it's a little bit deeper than that. These hierarchies can tend to be, you know, three levels deep, but you are constrained to these parent-child relations. You're not going further than that, and this is where we really begin to encode context and meaning.

The reason why this is critical is that when you introduce, and I'm going to point back to SCOSS and the ANSI Z39 standard, is that when we introduce parent-child relations in that level of context is that we actually are beginning to shape inference. Inference is important because we can assume if A, if B is part of A, then C is part of A, so we're starting to become a little bit more semantically expressive. We're starting to embed inference within our structure, which sets us up for having that richer context.

So, taxonomy being hierarchical, we have our concepts from our controlled vocabulary. It may have been a flat list, and then we're describing the concepts, and we're structuring them in parent-child relations. There are different types of hierarchical classification systems.

Primarily, we're looking at hierarchies and polyhierarchies, and a polyhierarchy, not to get too much into the weeds, is one concept may have two parents or three parents within the same taxonomy. Those become more complex and more difficult to manage or maintain over time, but I do want to note that a taxonomy is also any of these structures, but a taxonomy is also a living and breathing thing, so we're having to constantly maintain, introduce, and incorporate new concepts as they enter the system, refine, curate, and it's this repetitive cycle. It's not a one-and-done.

Many ask, is this something that an AI system can generate? Not very well. I actually have an article coming out soon about this where I test it across multiple models, so this is something that needs human touch. It's really human-AI collaboration, and it starts to get into a little bit more complexity to manage this at scale.

So, with the taxonomy that provides more disambiguation of concepts, there are multiple uses for a taxonomy. You can see them being used as decision trees. You can see them being used in order to help drive accuracy and categorization for vectorization for vector databases.

That's one implementation. For clustering, that's another implementation, and for helping to expand context so that we're able to incorporate and map to other taxonomies within an organization so we create this sort of network effect when needed, if needed, for AI implementations. One of the most important things is the human aspect, and I've touched on this quite a bit, but this is something I built at Adobe, and the idea is you have a taxonomy.

It's well-defined. It's encoded using SCOSS, which is great. You may have some custom class labels to extend the description, but the idea is that you have technical and non-technical people that are engaging with this taxonomy, and so creating an internal app that's very, very simple, sort of a glossary-style or catalog-style app, where people can actually align and see the vocabulary, whether they be in marketing or deeply technical working in DevOps or whatever, and that we can all agree upon the same definition, same shape, the same form, and then allowing people to give feedback.

That's really, really important because a lot of knowledge is locked within organizations and is, in fact, tacit, not explicit, so we need to start to collect that tacit knowledge. This lines up with what's called process engineering, is documenting how things are done within an organization, so this is especially important when we're designing AI systems for workflows, as an example, is that tacit knowledge and understanding how things are done.